TO ANALYSE SALES PERFORMANCE BY CATEGORY

1. Project Overview and Objective:

To analyse sales data and convert raw business data into meaningful insights using interactive dashboards. The project focuses on understanding overall sales performance, profit trends, customer behaviour, and category-wise performance to support data-driven decision-making.

Using Excel and Power BI, data is cleaned, modelled, and visualized to help stakeholders quickly identify key metrics such as total sales, profit, quantity sold, and target achievement across different dimensions like country, category, sub-category, and segment.

The main objective is to demonstrate data pre-processing techniques using Excel and an interactive Power BI dashboard visualization to make informed decisions.

2. Data Sources:

Source Description and Timeline: https://learn.microsoft.com

Domain : Finance / Sales Analytics etc

3. Problem Statement:

To analyse Which month and year had the most profit sales performance across regions and products to identify top-performing areas and profit trends.

To study, where is the company seeing the most success (by country/region)?

To evaluate which product and segment should the company continue to invest in?

To understand customer purchasing behaviour to enhance marketing and retention strategies.

4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description

Segment	String (Text)	Unique identifier for each segment
Category	String (Text)	Product Category
Product	String (Text)	Products
Manufacturing Price	Numeric (Integer)	Price for Each Products
Unit Sold	Numeric (Integer)	Unit Values
Discount	Numeric (Integer)	Discount for each Products
Sales Price	Numeric (Integer)	Sales Price for each Products
Date and Month	Numeric (Integer)	Month and Date
Year	Numeric (Integer)	Year
Profit	Numeric (Integer)	Profit amount by category

---

5. Tools & Technologies :

Excel : Data cleaning, transformation, and Pivot Tables.

Power BI : Data modelling, DAX calculations, visualization, and interactive dashboard creation.

---

6. Data Pre-Processing (Excel / Power Query):

Tasks Performed:

Data Cleaning & Transformation : Removed duplicates, handled missing values.

1. Replace blank Discount Band → "No Discount"
2. Ensure Date → Date format
3. Create Calendar Table (Power BI best practice)
4. Rename columns (remove extra spaces)
5. Sales and Manufacturing prices converted into decimal numbers.
6. Using conditional formatting the highest sales and higher profit values are highlighted.

Filtering & Sorting : Organized data to focus on relevant records.

Pivot Tables : Generated Pivot Tables for data summarisation and initial insights.

Convert the data into Fact and Dimension Table.

Fact Table :

Financial Data

Dimension Tables :

Date

Product

Country

Segment

Discount Band

7. Data Modelling and DAX (Power BI)

Data Model :

DAX MEASURES :

Total Sales = SUM('Financial'[Sales])

Total Profit = SUM('Financial'[Profit])

Total Units Sold = SUM('Financial'[Units Sold])

Profit Margin % =DIVIDE([Total Profit], [Total Sales], 0)

Total Discounts = SUM('Financial'[Discounts])

SCREENSHOT OF DATA MODELLING :

Calculated Columns & DAX Measures: Implemented DAX formulas for key metrics, such as total sales, growth percentages, and cumulative totals. (List Out)

8. Analysis and Visualizations (Power BI)

Dashboard Features:

· Total Profit Trend by Year, Quarter & Month

· This chart shows

Monthly bars grouped by Quarter and Year (2022 & 2023)

Two measures:

Total Profit

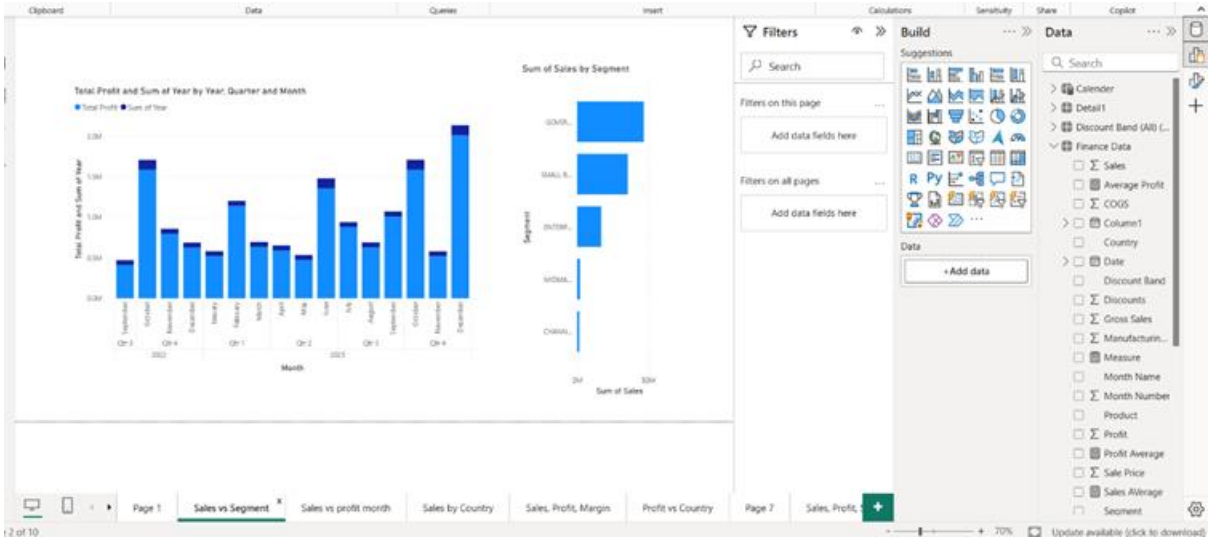
Sum of Sales (Yearly contribution)

Key Insights :

- Overall upward trend in 2023 compared to 2022, indicating business growth.
- Q4 (October–December) consistently shows higher profit, suggesting strong seasonal demand.
- December is the highest-performing month, with peak profit and sales.
- Mid-year months (April–June) show moderate performance, indicating relatively stable demand.
- Some dips in August and November, which may be due to:
    1. Higher discounts
    2. Increased costs
    3. Lower sales volume

This chart shows :

1. Sales distribution across different customer segments.
2. Segment Performance Ranking
3. Government Segment – Highest contributor to total sales
4. Small Business – Strong and consistent performance
5. Enterprise – Moderate contribution
6. Midmarket – Low sales volume
7. Channel Partners – Lowest contributor
8. Government and Small Business segments together drive the majority of revenue.
9. Channel Partners and Midmarket segments underperform, contributing minimal sales.
10. Enterprise segment shows potential for growth but currently lags behind top segments.



Total Profit by Month and Year – Descriptive Analysis :

This visual represents :

- This line chart displays the monthly total profit trend across different years, allowing us to identify:
- Profit fluctuations by month
- Seasonal patterns
- High-performing and low-performing periods
- The X-axis shows Month Name and Year, while the Y-axis represents Total Profit.

High-Performing Months :

- December 2022 records the highest profit, representing a peak business period.
- October 2023 also shows strong profit performance, confirming consistent year-end growth.
- June and July 2023 display moderate-to-high profits, indicating mid-year stability.

Low-Performing Months :

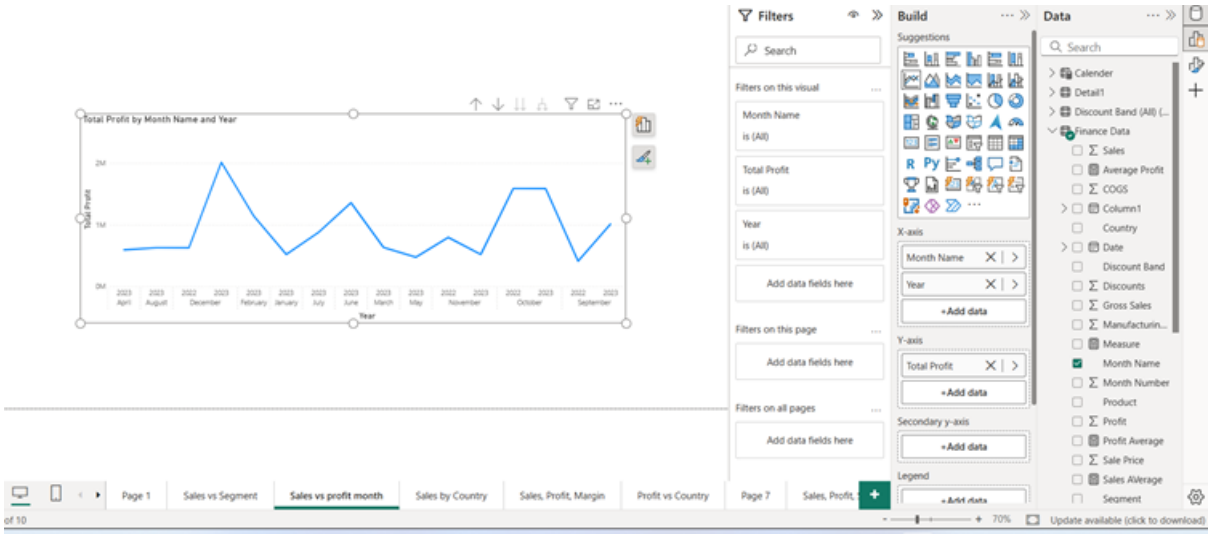
- September 2022 and January 2023 show lower profit values, possibly due to:
    - Reduced customer demand
    - Higher operational costs
    - Increased discounts or promotions

- These dips suggest periods where cost control and sales strategies need improvement.

Year-over-Year Observation :

- The profit trend in 2023 appears more consistent compared to 2022.
- Although peaks exist in both years, 2023 shows improved stability, indicating better financial management or optimized pricing strategies.

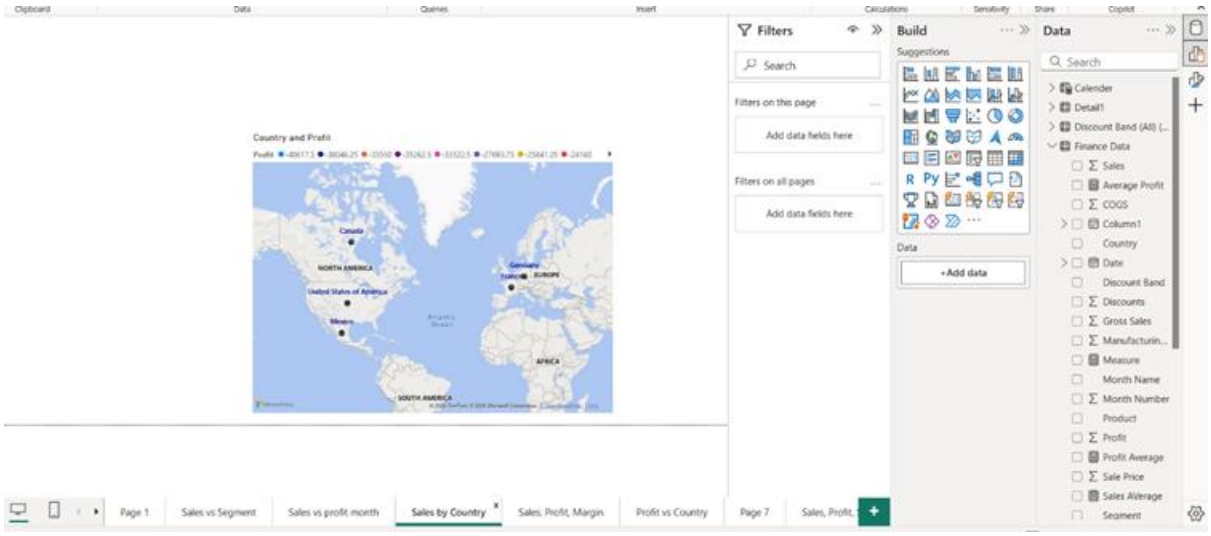
.



Sales by Country – Profit Insight (Map Visual)

· This map shows profit distribution by country. The analysis indicates that the United States and Canada contribute higher profit levels compared to other regions. European countries such as Germany and France show moderate profits, while Mexico records relatively lower profit. Overall, profitability is concentrated in North America, suggesting strong market performance in this region and potential growth opportunities in other countries.

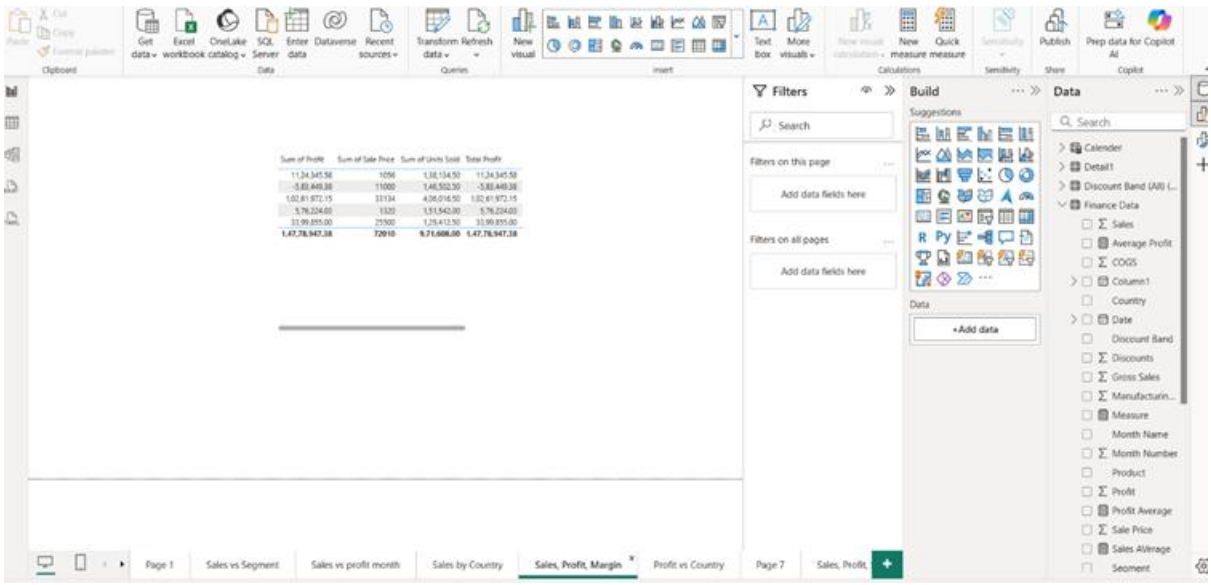
.



Sales, Profit & Margin – Summary Insight

This table provides an overall summary of sales price, units sold, and profit performance. While most categories generate positive profit, one segment shows a negative profit, indicating potential issues with pricing, discounts, or cost structure. The total figures highlight strong overall profitability, suggesting that high sales volume and effective pricing strategies are driving business performance, while loss-making areas require focused optimization.

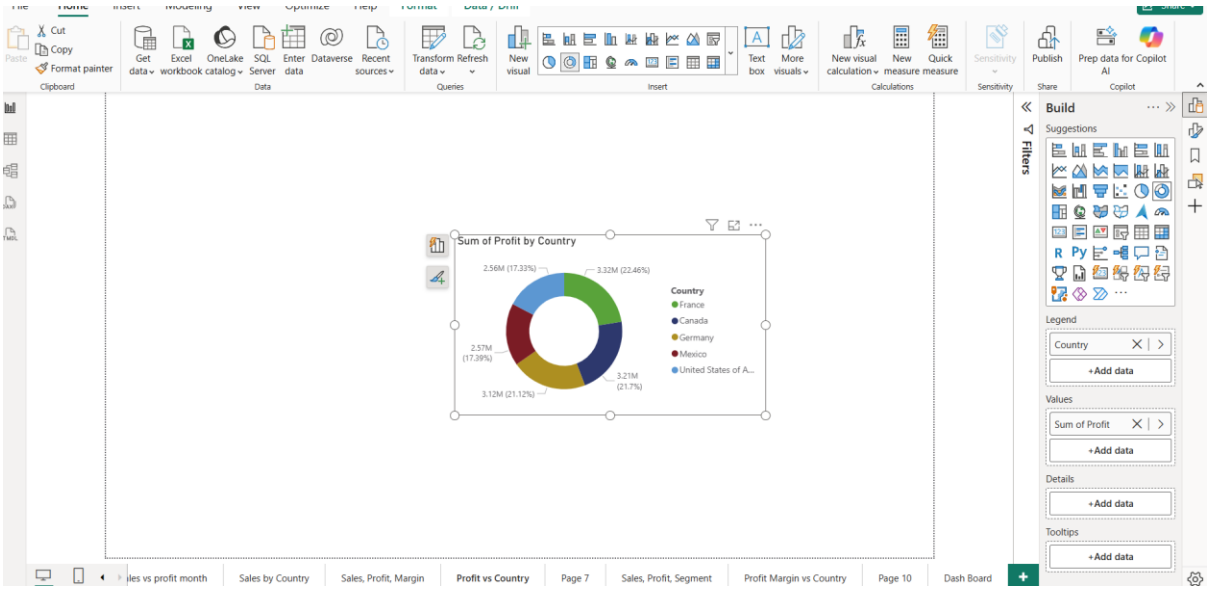
.



Profit Contribution by Country – Insight :

This chart shows the distribution of total profit across countries. France and Canada are the top contributors, each accounting for around 22% of total profit, followed closely by Germany. Mexico and the United States contribute a slightly smaller share. Overall, profits are fairly evenly distributed across regions, indicating balanced international performance with no single country overly dominating profitability.

"Profit is evenly distributed across countries, with France and Canada leading as the highest contributors."



Sales vs Profit by Segment – Insight :

This chart compares sales value and profit across customer segments. The analysis shows a positive relationship between sales and profit, where segments with higher sales generally generate higher profits. One segment stands out with high sales and high profit, making it the most valuable segment, while another segment shows lower sales and minimal or negative profit, indicating the need for pricing or cost optimization.

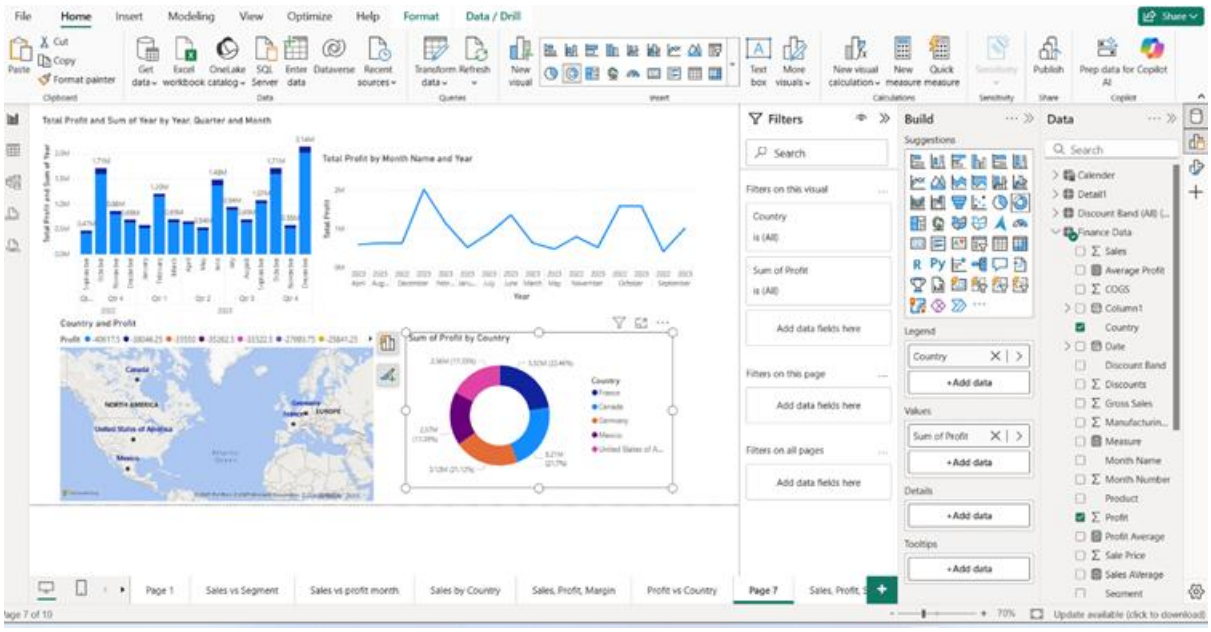
· "Higher sales segments drive higher profits, while low-performing segments require strategic review."

Profit

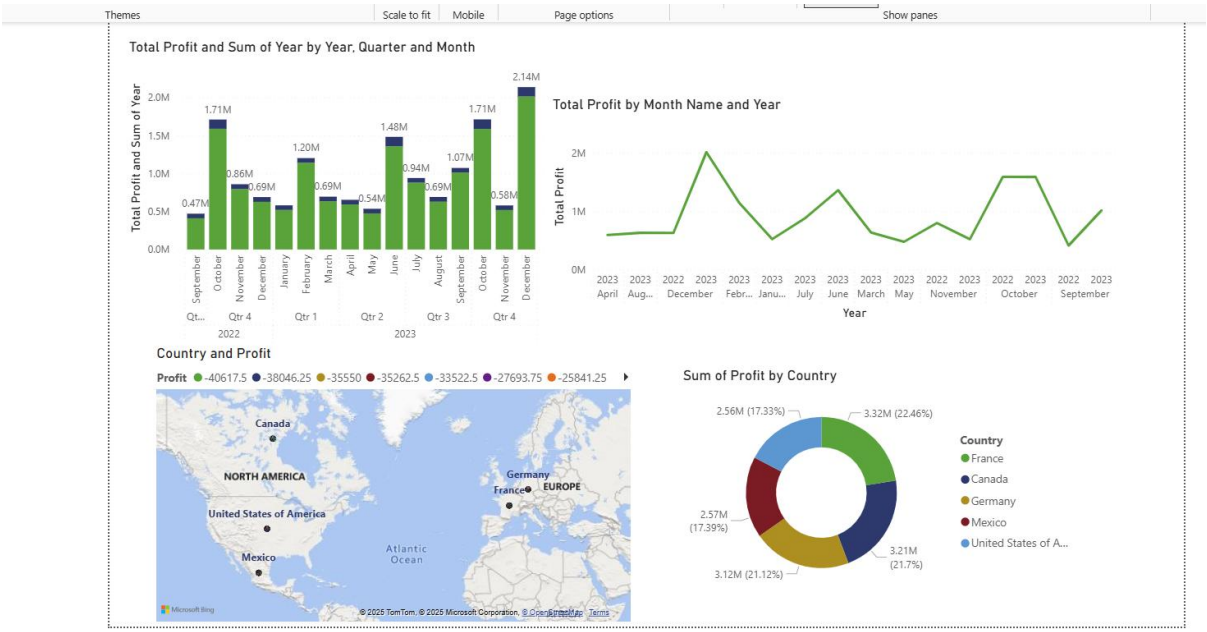
Performance Overview – Key Insights:

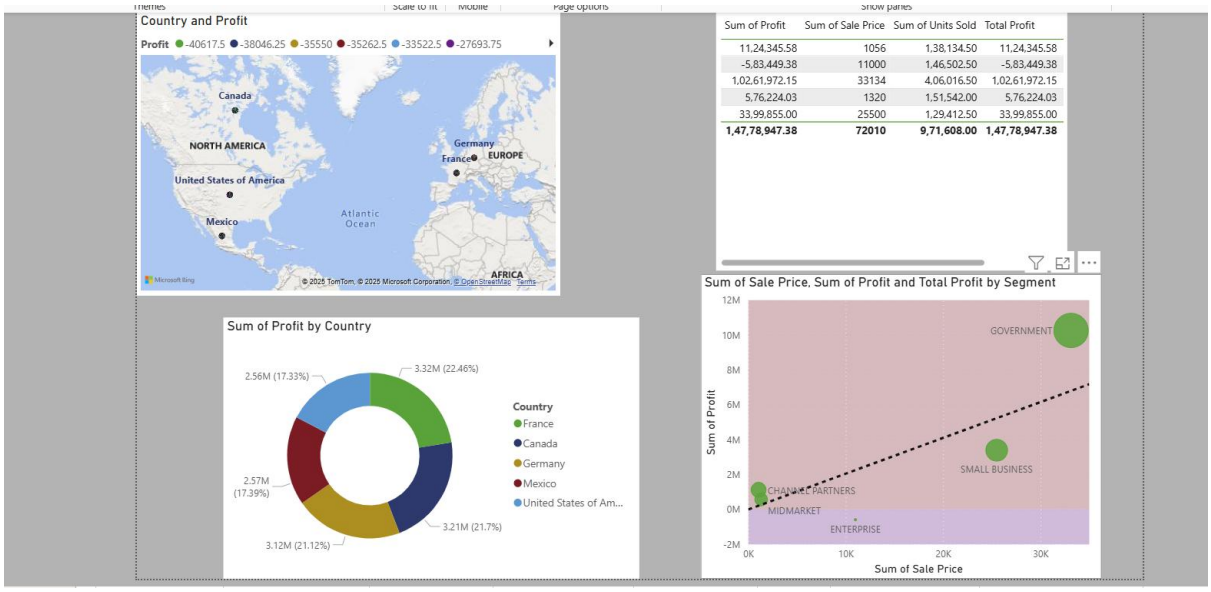
This dashboard highlights profit trends over time and across countries. Profits show a clear upward trend in 2023, with peak performance during Q4, indicating strong seasonal demand. Monthly analysis reveals December as the highest profit-generating month. Country-level insights show balanced profit contribution across regions, with France and Canada leading, while other countries contribute steadily. Overall, the business demonstrates consistent growth with diversified geographic profitability.

"Profit performance shows strong seasonal growth in 2023 with balanced contributions across countries."



SCREENSHOT OF DASHBOARD /REPORT :





Descriptive Insight – Sample Statements

1. **Overall Performance:**
   - *This analysis presents an overview of overall business performance by highlighting total sales, profit, and quantity trends across different periods and categories.*
2. **Sales Trend:**
   - *Sales show noticeable variation over time, with certain months contributing significantly higher revenue, indicating seasonal demand patterns.*
3. **Category-wise Performance:**
   - *Among all categories, a few key categories contribute the majority of total sales and profit, while others show relatively lower performance.*
4. **Profit Analysis:**
   - *Profit levels vary across categories and sub-categories, suggesting differences in cost structure and margin efficiency.*
5. **Top & Bottom Performers:**
   - *A small number of products and regions emerge as top performers, while some segments consistently underperform.*
6. **Regional Insights:**

- *Sales and profit distribution differs across regions, with certain regions driving a larger share of overall revenue.*

7. **Quantity vs Profit:**
   - *Higher sales quantity does not always result in higher profit, indicating the impact of pricing and discount strategies.*
8. **Customer Demand Pattern:**
   - *Customer purchasing behavior shows preference for specific product categories, reflected in higher order volume.*

This dashboard provides a comprehensive view of sales, profit, and performance trends across categories, regions, and time periods.

10. Conclusions :

The analysis shows strong and improving business performance, with profits increasing in 2023 compared to 2022. There is a clear seasonal pattern, where Q4 (especially December) consistently delivers the highest profits, indicating effective year-end demand and sales strategies. Profit contribution is well balanced across countries, with France and Canada leading while other regions provide steady support, reducing dependency on a single market. Customer segment analysis indicates that higher sales generally result in higher profits, though a few low-performing areas highlight opportunities for cost control and pricing optimization.

Overall, the business demonstrates healthy growth, stable geographic diversification, and strong seasonal performance, with scope to further improve profitability by optimizing underperforming months and segments.